

Plot analysis (GPT-4o)

The ternary plot illustrates the variation of the Peierls barrier in the Nb-Mo-Ta alloy system. Here's a detailed analysis:

Trend Identification

- **High Peierls Barrier:** The highest values (around 0.095 eV) are observed in compositions with a high percentage of Mo, particularly towards the Mo apex of the triangle. This suggests that Mo significantly contributes to increasing the Peierls barrier.
- **Low Peierls Barrier:** The lowest values (around 0.055 eV) are found near the Ta apex, indicating that Ta-rich compositions tend to have lower Peierls barriers.

Comparative Analysis

- **Nb Influence:** As the Nb content increases, the Peierls barrier generally decreases, especially when moving from Mo-rich to Nb-rich compositions.
- **Mo Influence:** Mo appears to have a strong positive influence on the Peierls barrier, as evidenced by the gradient from the Mo apex towards the other two elements.
- **Ta Influence:** Ta tends to lower the Peierls barrier, as seen by the darker colors near the Ta apex.

Statistical Considerations

- The gradient of colors suggests a smooth transition in Peierls barrier values across the composition space, indicating consistent trends without abrupt changes.

Contextual Interpretation

- **Mechanical Properties:** A higher Peierls barrier typically indicates greater resistance to dislocation motion, suggesting that Mo-rich alloys might exhibit higher strength.
- **Applications:** Alloys with higher Mo content could be more suitable for applications requiring high strength, while Ta-rich alloys might be preferred where ductility is more critical.

This analysis provides insights into how varying the composition of Nb, Mo, and Ta can tailor the mechanical properties of the alloy for specific applications.

Figure S5: Results of the analysis performed by the multi-modal agent powered by GPT-4o for the plot shown in Figure 6(c)